FORM TP 2016055

TEST CODE **01212020**

MAY/JUNE 2016

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION

CHEMISTRY

Paper 02 – General Proficiency

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in TWO sections.
2. Answer ALL questions.
3. Write your answers in the spaces provided in this booklet.
4. Do NOT write in the margins.
5. Where appropriate, ALL WORKING MUST BE SHOWN in this booklet.
6. You may use a silent, non-programmable calculator to answer questions.
7. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
8. **If you use the extra page(s) you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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SECTION A

Answer ALL questions.

Write your answers in the spaces provided in this booklet.

Do NOT spend more than 30 minutes on Question 1.

1. (a) A group of students was required to carry out an experiment in order to determine the melting point of an unknown solid, **A**. The apparatus was set up as shown in Figure 1.

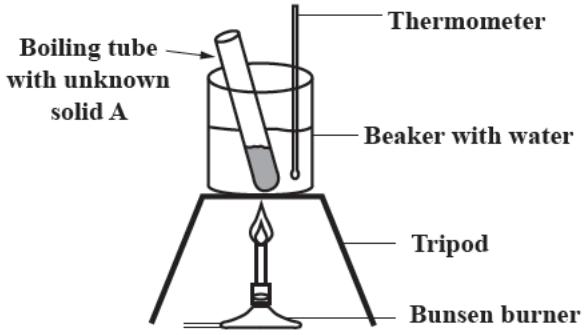

Figure 1. Diagram of apparatus set-up

Using the Bunsen burner, the water was heated to 80 °C. The Bunsen burner was turned off and temperature readings were recorded EVERY minute for 10 minutes.

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Figure 2 shows some of the thermometer readings recorded during the experiment.

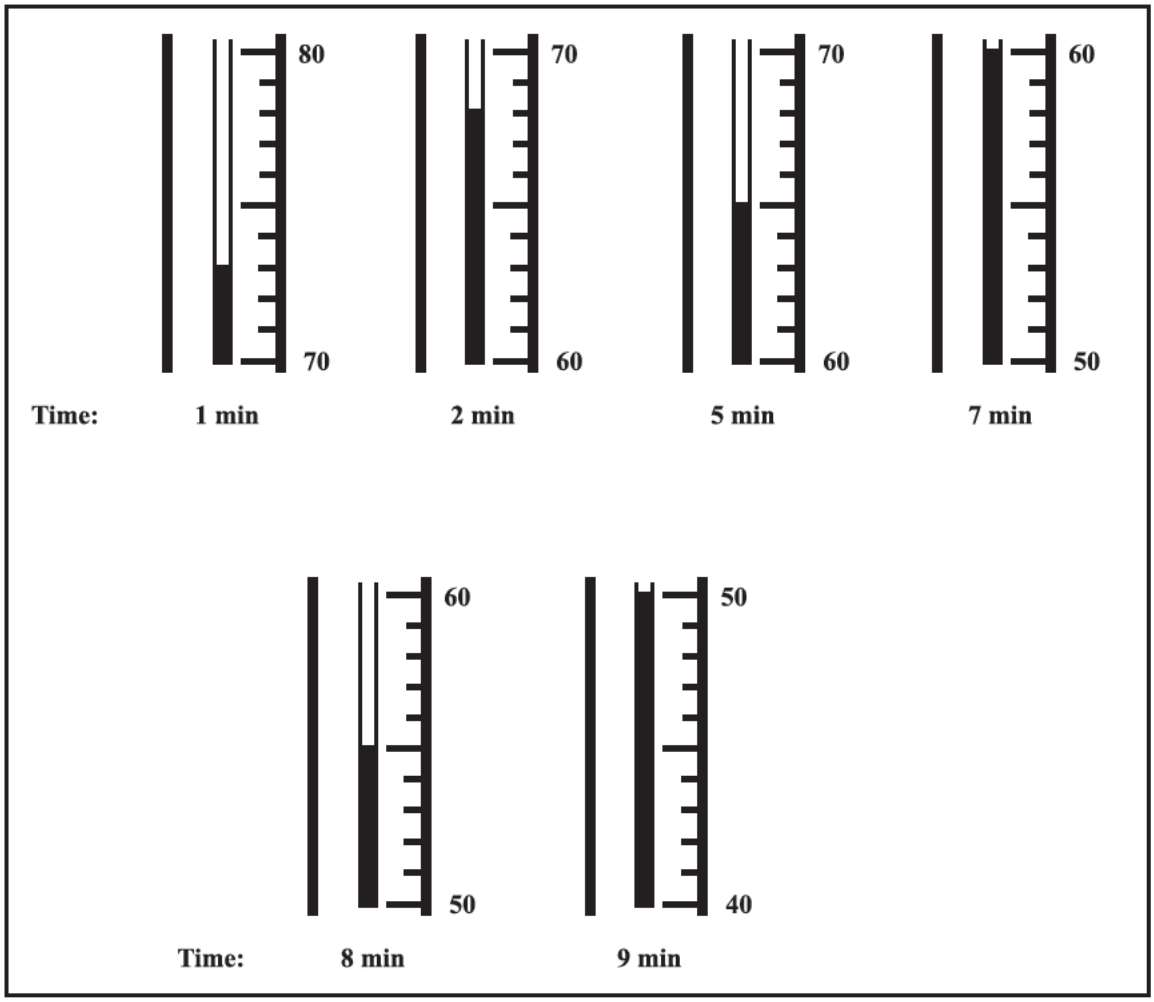

Figure 2. Thermometer readings

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(i) Complete Table 1 by inserting the thermometer readings from Figure 2 **(page 5)** at the appropriate times.

TABLE 1: RESULTS OF EXPERIMENT

Time (min)	**Temperature (°C)**
0	80
1	
2	
3	65
4	65
5	
6	64
7	
8	
9	
10	48

(4 marks)

(ii) Using the data from Table 1, plot the graph of temperature versus time on the grid provided in Figure 3 **on page 7**. Five points have already been plotted for you. Draw a smooth curve through the points to complete the cooling curve for **A**.

(5 marks)

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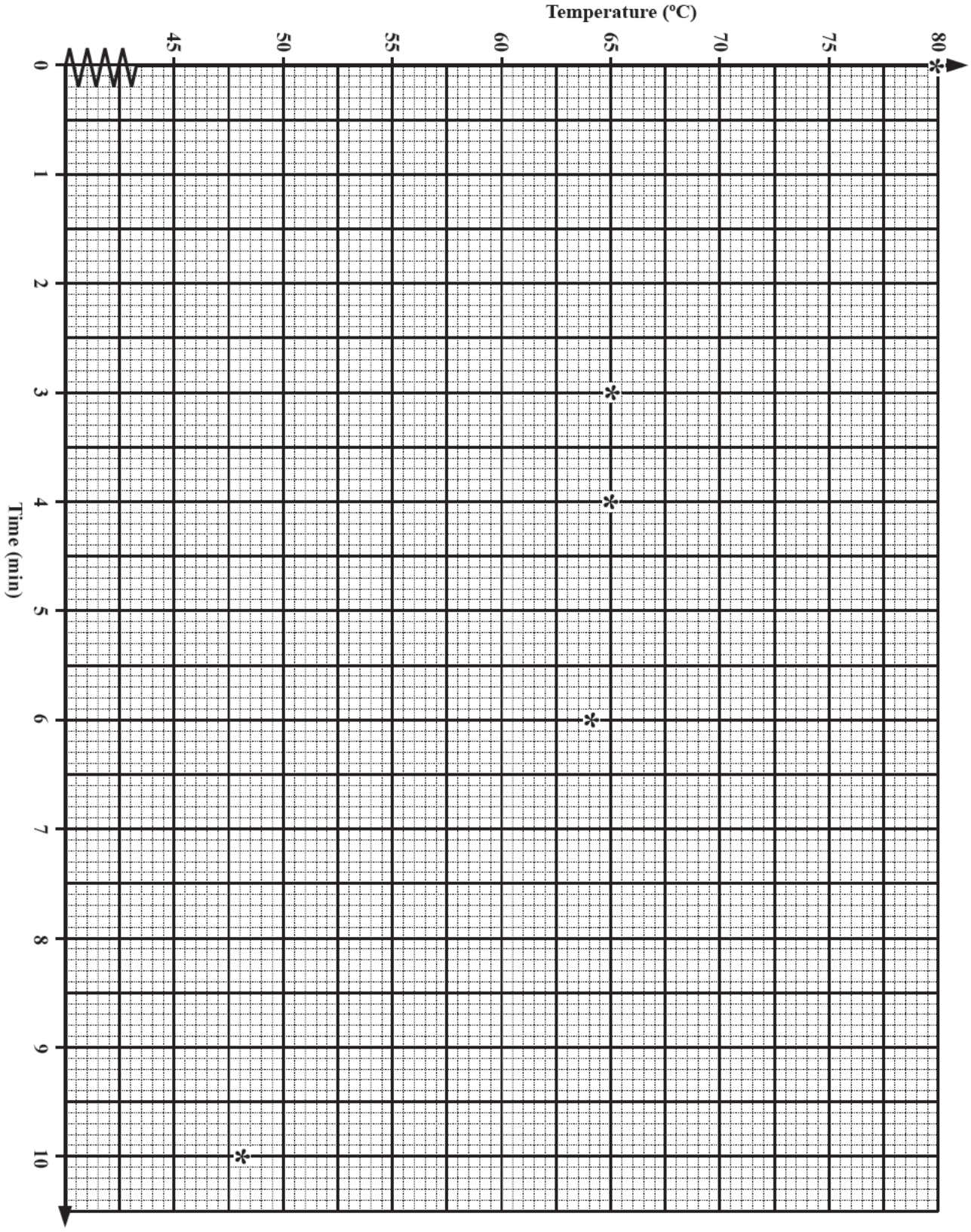

Figure 3. Cooling curve of A

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(iii) List the THREE states of matter.

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(1 mark)

(iv) Define the term 'melting point'.

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(2 marks)

(v) Based on information from the graph plotted in (a) (ii), state the melting point of **A**.

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(1 mark)

(vi) From the graph plotted in (a) (ii), identify the time range during which there should be a mixture of liquid and solid in the boiling tube.

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(1 mark)

(vii) State ONE precaution, other than wearing safety apparel, which should be taken during this experiment to avoid injury to the face or hands.

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(1 mark)

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(viii) Complete Figure 4 by drawing the arrangement of the particles of **A** at 0 min and 10 min. Use • (one dot) to represent a particle. **(2 marks)**

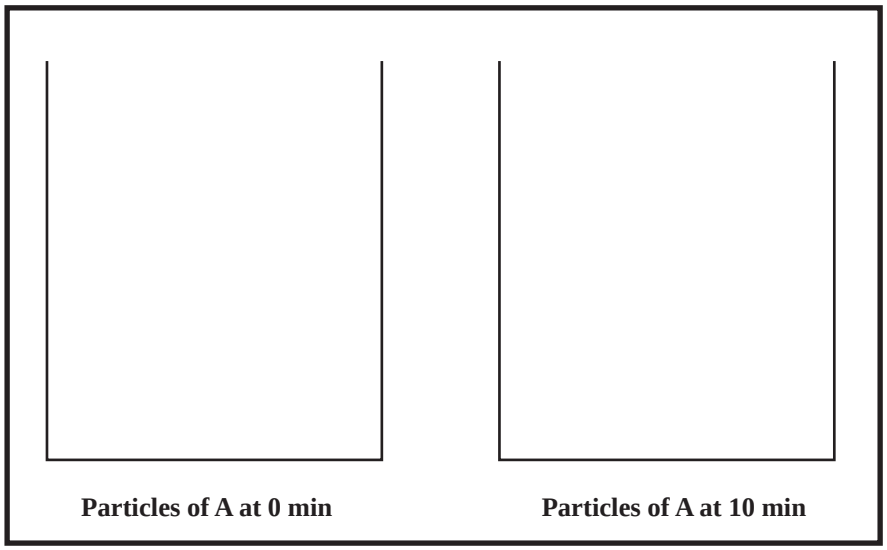

Figure 4. Particles of A

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(b) In another experiment, the group of students was given another unknown solid, **B**, and asked to identify it. They performed a series of tests and made some observations which are recorded in Table 2. Complete Table 2 by inserting the appropriate inferences.

(8 marks)

TABLE 2: RESULTS OF TESTS ON UNKNOWN SOLID, B

	Test	Observation	Inference
(i)	Place a small portion of solid, **B** in a test tube and heat strongly using a Bunsen flame.	• A reddish brown gas evolved	• **(1 mark)**
(ii)	Dissolve the remainder of **B** in about 10 cm^3 of distilled water, stir, then filter. Collect the filtrate and divide it into three equal portions for use in the following tests.		
(iii)	To the first portion of the filtrate from (b) (ii), add NaOH solution slowly until in excess.	• White precipitate formed • Soluble in excess	• • **(2 marks)**
(iv)	To the second portion of the filtrate from (b) (ii), add aqueous NH_3 slowly until in excess.	• White precipitate formed • Insoluble in excess	• • **(2 marks)**
(v)	To the third portion of the filtrate from (b) (ii), add aqueous KI.	• Yellow precipitate formed	• (Ionic equation required) • **(3 marks)**

Total 25 marks

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NOTHING HAS BEEN OMITTED.

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2. (a) Distinguish between an 'oxidizing agent' and a 'reducing agent' in terms of electron gain or loss.

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(2 marks)

(b) Jaden carried out some tests and recorded the observations in Table 3 below. Use the information in the table to answer the questions that follow.

TABLE 3: OBSERVATIONS

Test		Observation
1	An aqueous solution of **C** was added to acidified potassium manganate(VII).	Potassium manganate(VII) solution changed from purple to pale pink.
2	An aqueous solution of **C** was added to a solution of potassium iodide.	Potassium iodide solution changed from colourless to brown.
3	A strip of iron was placed into an aqueous solution of copper(II) sulfate.	Blue solution faded and brown deposits settled.
4	A strip of silver was placed into an aqueous solution of copper(II) sulfate.	Solution remained blue.

(i) State an oxidizing agent that is NOT mentioned in Table 3.

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(1 mark)

(ii) Identify whether Solution **C** is acting as an oxidizing or reducing agent in Test 1 and Test 2.

Test 1:

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Test 2:

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(2 marks)

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(iii) With reference to oxidation states, explain EACH of your answers in (b) (ii).

Test 1:

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Test 2:

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(4 marks)

(iv) Write an ionic equation (with state symbols) for the reaction which took place in Test 3.

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(2 marks)

(v) With reference to the electrochemical series, explain the difference in observations between Tests 3 and 4.

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(4 marks)

Total 15 marks

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3. (a) Define the term 'structural isomerism'.

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(2 marks)

(b) Compound P, $H_2C{=}CHCH_2CH_2CH_3$, is a hydrocarbon.

(i) State the name of Compound P.

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(1 mark)

(ii) State ONE physical property of Compound P.

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(1 mark)

(iii) State ONE chemical property of Compound P.

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(1 mark)

(iv) Draw the FULLY DISPLAYED structural formulae of Compound P and ONE of its isomers.

Compound P	**Isomer of Compound P**

(4 marks)

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(c) Compound P reacts with hydrogen gas in the presence of a catalyst to form Compound Q.

(i) Draw the FULLY DISPLAYED structure of ONE BRANCHED isomer of Compound Q.

(2 marks)

(ii) State ONE physical property of Compound Q.

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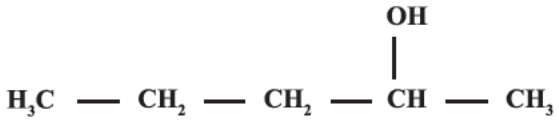
(1 mark)

(d) The following diagram shows the structure of Compound R.



Compound R

State the name of Compound R.

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(1 mark)

(e) State the conditions required for the dehydration of ethanol.

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(2 marks)

Total 15 marks

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SECTION B

Answer ALL questions.

4. (a) Graphite and diamond are two allotropes of carbon. Define the term 'allotrope'. Illustrate your answer by drawing the lattice structures for **both** diamond and graphite.

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Diamond	**Graphite**

(6 marks)

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(b) Like metals, graphite can conduct electricity but diamond cannot conduct electricity.

(i) Describe the bonding present in metals.

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(2 marks)

(ii) Explain, in terms of bonding, why graphite conducts electricity but diamond does NOT conduct electricity.

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(2 marks)

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(c) Sodium chloride and calcium oxide both have similar giant ionic crystalline structures with high melting points. However, the melting point of calcium oxide is higher than that of sodium chloride. Sketch the lattice structure for sodium chloride and suggest why its melting point would be lower than that of calcium oxide.

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(5 marks)

Total 15 marks

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5. (a) Methane, CH_4, is the simplest hydrocarbon known to man.

Identify the type of bonding present in methane. State a reason for your answer.

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(3 marks)

(b) Methane, like other alkanes, is generally unreactive. However, under certain conditions, it can undergo reactions with chlorine, Cl_2, forming carbon tetrachloride, CCl_4.

(i) Write a balanced equation for this reaction.

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(2 marks)

(ii) State the type of reaction taking place and identify the reaction conditions.

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(2 marks)

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(c) Identify and draw the structural formulae of the alkene and the acid from the list of molecular formulae given below.

C_4H_8, C_4H_{10}, $C_4H_8O_2$, $C_4H_{10}O$

Alkene: _____ **Acid:** _____

(4 marks)

(d) For EACH of the following pairs of compounds, identify a test that can be used to distinguish between them. In your answer, include the expected observation(s).

(i) An alkane and an alkene

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(2 marks)

(ii) An alkane and a carboxylic acid.

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(2 marks)

Total 15 marks

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6. (a) The Caribbean islands are known for their many resources. For example, the ore of aluminium is found in Jamaica and Guyana.

(i) Briefly describe the extraction of aluminium from its named ore. Include in your description ONE ionic equation.

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(5 marks)

(ii) Although aluminium ore is mined in the Caribbean, the extraction of aluminium usually takes place in other countries. Suggest ONE reason why it may be difficult to extract aluminium in the Caribbean.

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(2 marks)

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(b) An alloy can be described as a 'mixed metal solution'. Aluminium forms an alloy with magnesium that is used to make aircraft.

Identify this alloy and suggest why it is better to use it, rather than the pure metal, in the manufacture of aircraft.

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(5 marks)

(c) "Some metals can be useful to man but yet harmful to the environment." Comment on this statement with reference to mercury.

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(3 marks)

Total 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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